Linux Filesystem

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Review Questions

Q1) Difference between 6 and 9 degrees of freedom imu.

Ans) A 6 DoF IMU has an accelerometer and a gyroscope but a 9 DoF IMU has a magnetometer also. This magnetometer adds 3 more DoF to the IMU other than the DoF of accelerometer and gyroscope. A magnetometer is used to measure strength of magnetic field and acts like a compass.

Q2) What are the 3 degrees of freedom wrt magnetic component of imu?

Ans) The 3 DoF with respect to magnetic component of IMU are measured along X,Y and Z axis. The X-axis DoF measures forward and backward magnetic field. The Y-axis DoF measures the left and right magnetic field. Together the X and Y axis are used to find the heading and directions for the rover. The Z-axis DoF measures the vertical magnetic field and is used to find inclination of magnetic field from the ground.

Q3) Encoder vs IMU what is better for use on the arm?

Ans) An encoder measures the angle between joints in the arm with respect to one another in the rover but the IMU measures the acceleration and rotational speed with respect to the ground. This makes the encoder more accurate as the IMU calculates the position by estimating the time a rotation must be made which can cause a drift error over time.

Q4) Why is stereo camera digital?

Ans) A stereo camera is digital because it uses electronic image sensors and processors to capture, align, and compute depth from two separate views instantly. Digital processors measure exact distances by matching identical pixels between the left and right image sensors. Digital systems build 3D depth maps, which is too slow with traditional film.

Q5) Git reset vs git revert?

Ans) Git reset shifts the current branch head to an older commit, by erasing the newer commits. Git revert acts as a safety net. It calculates the opposite of a specified commit and applies it as a brand-new commit. The broken commit still exists in the project log, but a new commit immediately updates its code modifications.

Q6) Git pull vs git fetch?

Ans) Git pull automatically merges remote changes into your local working files, whereas git fetch only downloads the changes into your local git database without changing your code. Git pull is a shortcut for running git fetch followed immediately by git merge. Git fetch gives you complete control and ensures you are never blindsided by unexpected merge conflicts. Git pull is used when you want a quick sync.

Q7) How many commands in git config?

Ans) The git config command has 7 built-in subcommands which are list, get, set, unset, rename-section, remove-section and edit.

list: Lists all variables set in configuration files along with their values.

get: Retrieves the value of one or more configuration keys.

set: Sets the value for one or more configuration keys.

unset: Removes a configuration key matching the given option.

rename-section: Renames the given section to a new name.

remove-section: Removes the given section from the configuration file.

edit: Opens an editor to manually modify the specified configuration file.

Introduction to Linux Filesystem?

A Linux filesystem is the method used by the Linux operating system to organize, store, and manage our data on storage devices like hard drives and USB drives. It provides a hierarchical structure where files and directories are arranged in a tree-like format starting from a root directory. It's objectives are to store and retrieve data efficiently, organize files and directories systematically, manage permissions and ownership, support multiple users and processes and ensure data security and integrity.

Key Features of Linux Filesystem

Hierarchical Directory Structure: Linux uses a tree-like structure with the root directory at the top. All files and directories branch out from this root.

Everything is a File: In Linux, devices, directories, and processes are treated as files, making system management consistent and flexible.

Multi-User Support: Multiple users can access the system simultaneously while maintaining separate permissions and ownership.

Security and Permissions: Linux provides read (r), write (w), and execute (x) permissions for users, groups, and others.

Case Sensitivity: File names are case-sensitive

Support for Multiple Filesystems: Linux supports various filesystem types such as Ext4, XFS, Btrfs, FAT32, NTFS

Journaling: Modern Linux filesystems maintain a journal that helps recover data after unexpected system crashes.

Linux Directory Structure

The Linux filesystem begins with the root directory.

Directory	Description
/	Root directory containing all files and directories
/bin	Essential user commands and executable files
/boot	Boot loader files and kernel
/dev	Device files
/etc	System configuration files
/home	User home directories
/lib	Shared libraries required by system programs
/media	Mount point for removable media
/mnt	Temporary mount point
/opt	Optional software packages
/proc	Virtual filesystem containing process information
/root	Home directory of the root user
/sbin	System administration commands
/tmp	Temporary files
/usr	User applications and utilities
/var	Variable data such as logs and mail files

Basic Linux Commands

pwd (Print Working Directory): Displays the current directory.

ls (List Files and Directories): Shows files and folders in a directory.

cd (Change Directory): Moves between directories.

mkdir (Make Directory): Creates a new directory.

rmdir (Remove Directory): Removes an empty directory.

touch: Creates an empty file.

cp (Copy Files): Copies files or directories.

mv (Move/Rename Files): Moves or renames files.

rm (Remove Files): Deletes files or directories.

cat: Displays file contents.

echo: Displays text or writes content to a file.

grep: Searches for patterns in files.

man: Displays the manual page of a command.

clear: Clears the terminal screen.